
Pin Map

Pin Name	Pin	Pin Name	Pin
OE	1	1Q	2
1D	3	2D	4
2Q	5	3Q	6
3D	7	4D	8
4Q	9	CLK	11
5Q	12	5D	13
6D	14	6Q	15
7Q	16	7D	17
8D	18	8Q	19

Global Parameters

Parameter	Value
VCC Pin	20
GND Pin	10
VCC Voltage	2.5, 3.3, 4.0, 4.5, 5
Input Low	0.75, 0.99, 1.2, 1.35, 1.5
Input High	1.75, 2.31, 2.8, 3.15, 3.5
Output Low	0.3, 0.3, 0.3, 0.3, 0.3
Output High	2.2, 3.0, 3.6, 4.1, 4.7

Tests: **IC PASS 16/16**

Flip-Flop 1: **PASS**

Inputs			VCC	Outputs/Results	
OE	1D	CLK		1Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 1 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	1D	CLK		1Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	H	L	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)

Flip-Flop 2: PASS

Inputs			VCC	Outputs/Results	
OE	2D	CLK		2Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.8 (H)
			4.5		4.4 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 2 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	2D	CLK		2Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.8 (H)
			4.5		4.41 (H)
			5		4.85 (H)
L	H	L	2.5	H	2.36 (H)
			3.3		3.15 (H)
			4.0		3.8 (H)
			4.5		4.41 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)

Flip-Flop 3: PASS

Inputs			VCC	Outputs/Results	
OE	3D	CLK		3Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 3 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	3D	CLK		3Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.86 (H)
L	H	L	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.86 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)

Flip-Flop 4: PASS

Inputs			VCC	Outputs/Results	
OE	4D	CLK		4Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 4 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	4D	CLK		4Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	H	L	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)

Flip-Flop 5: PASS

Inputs			VCC	Outputs/Results	
OE	5D	CLK		5Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 5 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	5D	CLK		5Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.85 (H)
L	H	L	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)

Flip-Flop 6: PASS

Inputs			VCC	Outputs/Results	
OE	6D	CLK		6Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 6 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	6D	CLK		6Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	H	L	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)

Flip-Flop 7: PASS

Inputs			VCC	Outputs/Results	
OE	7D	CLK		7Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 7 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	7D	CLK		7Q	
L	H	R	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	H	L	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.43 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)

Flip-Flop 8: PASS

Inputs			VCC	Outputs/Results	
OE	8D	CLK		8Q	
L	H	R	2.5	H	2.36 (H)
			3.3		3.15 (H)
			4.0		3.8 (H)
			4.5		4.41 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
H	X	X	2.5	Z	1.59 (Z)
			3.3		1.59 (Z)
			4.0		1.59 (Z)
			4.5		1.59 (Z)
			5		1.59 (Z)

Flip-Flop 8 No Change: PASS

Inputs			VCC	Outputs/Results	
OE	8D	CLK		8Q	
L	H	R	2.5	H	2.36 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.85 (H)
L	H	L	2.5	H	2.37 (H)
			3.3		3.15 (H)
			4.0		3.81 (H)
			4.5		4.42 (H)
			5		4.85 (H)
L	L	R	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)
L	L	L	2.5	L	0.0 (L)
			3.3		0.0 (L)
			4.0		0.0 (L)
			4.5		0.0 (L)
			5		0.0 (L)